```
  GNU nano 8.7.1
#include <stdio.h>
int main()
{
char str[100];
int start, length, i, count = 0;
printf("Enter a string: ");
fgets(str, 100, stdin);
while (str[count] != '\0' && str[count] != '\n')
{
count++;
}
printf("Length of string: %d\n",count);
printf("Enter starting position: ");
scanf("%d",&start);
printf("Enter number of characters: ");
scanf("%d", &length);
printf("Substring: ");
for (i = start; i < start+length && i< count; i++)
{
printf("%c", str[i]);
}
printf("\n");
return 0;
}
```

```
  GNU nano 8.7.1
#!/bin/bash
read -p "Enter a string: " str
echo "Length of string: ${#str}"
read -p "Enter starting position: " start
read -p "Enter number of characters: " length
echo "Substring: ${str:$start:$length}"
```

```
vboxuser@Kartik:~$ nano exp5.sh
vboxuser@Kartik:~$ bash exp5.sh
Enter a string: hello world
Length of string: 11
Enter starting position: 2
Enter number of characters: 5
Substring: llo w
vboxuser@Kartik:~$ time bash exp5.sh
Enter a string: hello world
Length of string: 11
Enter starting position: 2
Enter number of characters: 5
Substring: llo w

real    0m15.319s
user    0m0.002s
sys     0m0.002s
vboxuser@Kartik:~$
```

```
vboxuser@kartik:~$ nano exp5.c
vboxuser@kartik:~$ gcc exp5.c -o exp5
vboxuser@kartik:~$ chmod +x exp5
vboxuser@kartik:~$ ./exp5
Enter a string: hello world
Length of string: 11
Enter starting position: 2
Enter number of characters: 5
Substring: llo w
vboxuser@kartik:~$ time bash exp5.c
exp5.c: line 2: syntax error near unexpected token `('
exp5.c: line 2: `int main()'

real    0m0.004s
user    0m0.001s
sys     0m0.003s
```